

❖ **Brownian Motion 1 - Introduction**

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Historical Motivation

Brownian motion originates from a physical phenomenon observed in 1827 by the botanist **Robert Brown**. He noticed that tiny particles suspended in water move in an erratic, continuous, and seemingly random fashion. [Wikipedia](#)

In 1905, **Albert Einstein** connected Brownian motion with the **heat equation**

The density of particles at position x and time t is $f(x, t)$ satisfies the heat equation:

$$\frac{\partial}{\partial t} f(x, t) = \frac{1}{2} \frac{\partial^2}{\partial x^2} f(x, t).$$

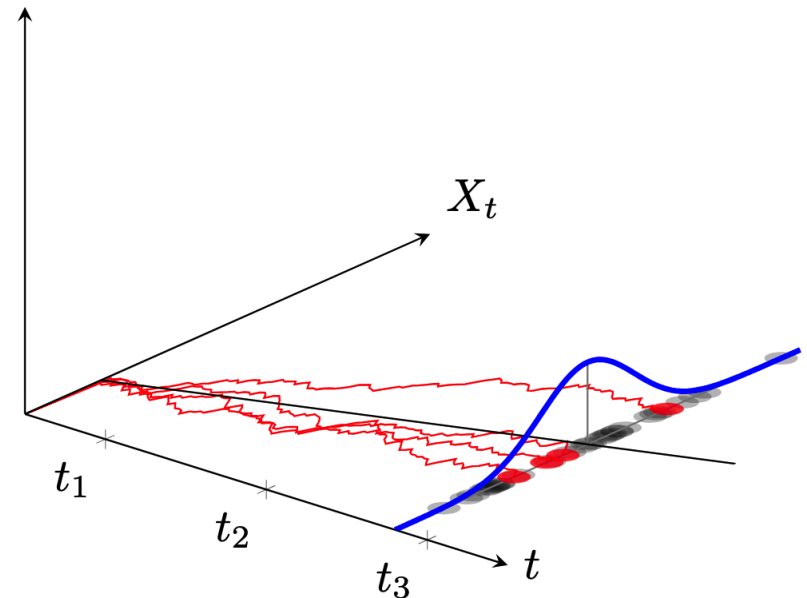
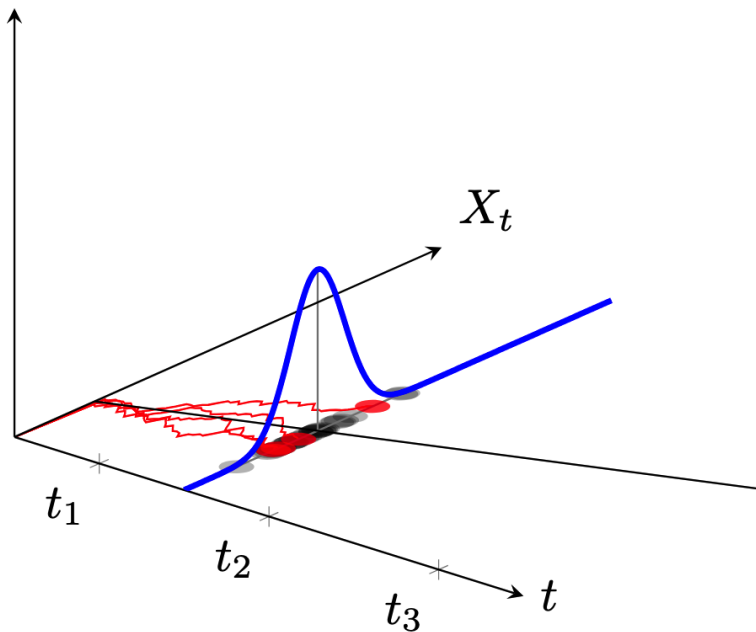
With initial condition $f(x, 0) = \delta(x)$, the solution is

$$f(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}.$$

This is exactly the density of a normal random variable: $X_t \sim \mathcal{N}(0, t)$.

This is the fundamental probabilistic signature of Brownian motion.

- The position of a diffusing particle at time t is normally distributed.
- Mean = 0,
- Variance = t ,
- Spread increases linearly in time.



Definition: Standard Brownian Motion

A continuous-time stochastic process $(B_t)_{t \geq 0}$ is called a **standard Brownian motion** (or Wiener process) if it satisfies:

1. Start at zero

$$B_0 = 0.$$

2. Normal distribution For every $t > 0$,

$$B_t \sim \mathcal{N}(0, t).$$

3. Stationary increments For all $s, t > 0$,

$$B_{t+s} - B_s \stackrel{d}{=} B_t.$$

That is, increments depend only on the time length, not on the starting time.

4. Independent increments

For $0 \leq q < r \leq s < t$, the increments $B_t - B_s$ and $B_r - B_q$ are independent random variables.

5. Continuous paths. With probability 1, the function $t \mapsto B_t$ is continuous.

Joint pdf

Assume $(B_t)_{t \geq 0}$ is standard Brownian motion and let $0 \leq s < t$.

Compute the covariance $\text{Cov}(B_s, B_t)$.

Write $B_t = B_s + (B_t - B_s)$.

$$\begin{aligned} E[B_s B_t] &= E[B_s(B_s + (B_t - B_s))] \\ &= E[B_s^2] + E[B_s(B_t - B_s)] \\ &= E[B_s^2] = s \end{aligned}$$

So the covariance matrix of the vector (B_s, B_t) is $K(s, t) = \begin{pmatrix} s & s \\ s & t \end{pmatrix}$.

The joint density is

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{1}{2\pi\sqrt{\det K}} \exp\left(-\frac{1}{2}(x \ y)K^{-1}\begin{pmatrix} x \\ y \end{pmatrix}\right) \\ &= \frac{1}{2\pi\sqrt{s(t-s)}} \exp\left(-\frac{1}{2s(t-s)}(tx^2 - 2sxy + sy^2)\right) \end{aligned}$$

Conditional and joint pdf

Since $B_t = B_s + (B_t - B_s)$ and $B_t - B_s$ is independent of B_s , conditioning on $B_s = y$ simply shifts the mean:

$$B_t \mid (B_s = x) \sim \mathcal{N}(x, t - s).$$

Therefore the conditional density is

$$f_{B(t)|B(s)}(y \mid x) = \frac{1}{\sqrt{2\pi(t-s)}} \exp\left(-\frac{(y-x)^2}{2(t-s)}\right)$$

By the chain rule of densities,

$$f_{B_s, B_t}(x, y) = f_{B_t|B_s}(y \mid x) f_{B_s}(x).$$

$$= \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(y-x)^2}{2(t-s)}} \cdot \frac{1}{\sqrt{2\pi s}} e^{-\frac{x^2}{2s}}$$

Conditional and joint pdf

In the general finite-dimensional distribution.

Let

$$0 = t_0 < t_1 < t_2 < \dots < t_n, \quad x_0 = 0.$$

Write increments:

$$B(t_j) = B(t_{j-1}) + (B(t_j) - B(t_{j-1})).$$

So we obtain:

$$\begin{aligned} f_{B(t_1), \dots, B(t_n)}(x_1, \dots, x_n) &= \prod_{j=1}^n f_{B(t_j) - B(t_{j-1})}(x_j - x_{j-1}) \\ &= \prod_{j=1}^n \frac{1}{\sqrt{2\pi(t_j - t_{j-1})}} \exp\left(-\frac{(x_j - x_{j-1})^2}{2(t_j - t_{j-1})}\right) \end{aligned}$$

with $t_0 = 0$ and $x_0 = 0$

Brownian motion is completely characterized by independent Gaussian increments.

Brownian Motion as the Limit of a Random Walk

Let N be a positive integer. Consider a standard symmetric random walk

$$Z_0, Z_1, Z_2, \dots, Z_{2^N},$$

where

$$Z_0 = 0, Z_k = \sum_{i=1}^k \xi_i,$$

and the increments ξ_i are independent with

$$P(\xi_i = 1) = P(\xi_i = -1) = \frac{1}{2}.$$

Thus, after k steps, Z_k takes values in

$$\{-k, -k + 2, \dots, k - 2, k\}$$

We want to build from this a continuous-time process defined on the interval $0 \leq t \leq 1$. To do this, we rescale both time and space.

For $0 \leq t \leq 1$, define

$$X_t^{(N)} = 2^{-N/2} Z_{\lfloor 2^N t \rfloor},$$

where $\lfloor 2^N t \rfloor$ denotes the integer part of $2^N t$.

This construction has a very clear meaning:

- Time is divided into 2^N equal subintervals of length 2^{-N} .
- At each such time step, the random walk jumps by ± 1 .
- We speed up time by a factor 2^N .
- We shrink space by a factor $2^{N/2}$.

So each jump of the original walk, which has size ± 1 , becomes a jump of size $\pm 2^{-N/2}$ in the scaled process.

- The time between jumps is 2^{-N} , so jumps happen more and more frequently,
- The size of each jump is $2^{-N/2}$, so each jump becomes smaller and smaller,
- The trajectory begins to resemble a continuous curve.

State Space

Since Z_k takes values in

$$\{-2^N, -2^N + 2, \dots, 2^N\},$$

the scaled process $X_t^{(N)}$ takes values in

$$\Omega_N = 2^{-N/2} \{-2^N, -2^N + 2, \dots, 2^N\},$$

Thus Ω_N is a discrete lattice in $\mathbb{R}$ whose spacing is

$$2^{-N/2} \cdot 2 = 2^{1-N/2}.$$

As $N \rightarrow \infty$, this lattice becomes dense in $\mathbb{R}$.

Why This Scaling?

The scaling is chosen so that the variance matches Brownian motion.

Recall: $\text{Var}(Z_k) = k$. Hence:

$$\text{Var}\left(X_t^{(N)}\right) = 2^{-N} \text{Var}\left(Z_{\lfloor 2^N t \rfloor}\right) = 2^{-N} \lfloor 2^N t \rfloor \approx t.$$

Piecewise Constant Trajectories

The process $X_t^{(N)}$ is constant on each interval

$$\left[\frac{k}{2^N}, \frac{k+1}{2^N}\right),$$

and jumps at the times

$$t = \frac{k}{2^N}, k = 0, 1, \dots, 2^N.$$

So:

- $X_t^{(N)}$ is a continuous-time, piecewise-constant process.
- It is right-continuous with left limits (a step function).
- Jumps occur at regular time intervals of length 2^{-N} .

This is a “polygonal” or “staircase” approximation to Brownian motion.

Discrete RW $\xrightarrow{\text{rescale time, space}}$ Continuous-time process $\xrightarrow{N \rightarrow \infty}$ Brownian motion.

Limit as $N \rightarrow \infty$

The goal is to show that as $N \rightarrow \infty$,

$$\{X_t^{(N)}: 0 \leq t \leq 1\} \Rightarrow \{B_t: 0 \leq t \leq 1\},$$

where B_t is standard Brownian motion.

This result is known as **Donsker's invariance principle**, or the **functional central limit theorem**.

one can construct the random walks and Brownian motion on the same probability space so that

$$\sup_{0 \leq t \leq 1} |X_t^{(N)} - B_t| \rightarrow 0 \text{ almost surely.}$$

This shows that Brownian motion has almost surely continuous paths and that the random walk approximations converge uniformly.

Transformations of Brownian Motion

Let $(B_t)_{t \geq 0}$ be a standard Brownian motion.

Fix $s \geq 0$ and $c \neq 0$, and define three new processes: Time-shifted increments, Scaling in time and space, and Time inversion:

$$A(t) = B(t + s) - B(s), \quad C(t) = c B\left(\frac{t}{c^2}\right), \quad D(t) = t B\left(\frac{1}{t}\right), \quad t > 0$$

Claim: Each of $A(t)$, $C(t)$, and $D(t)$ is a **standard Brownian motion**.

Since Brownian motion is a Gaussian process, any linear transformation of it is again Gaussian. So it suffices to check:

- Mean is zero,
- Covariance is $E[X(t_1)X(t_2)] = \min(t_1, t_2)$,
- $X(0) = 0$ and paths are continuous (which follow from continuity of B).

Proof for $C(t)$

$$\text{Mean: } E[C(t)] = c E \left[B \left(\frac{t}{c^2} \right) \right] = 0.$$

Covariance: for $0 < t_1 < t_2$,

$$E[C(t_1)C(t_2)] = c^2 E \left[B \left(\frac{t_1}{c^2} \right) B \left(\frac{t_2}{c^2} \right) \right].$$

Since Brownian covariance is $\min(u, v)$,

$$E \left[B \left(\frac{t_1}{c^2} \right) B \left(\frac{t_2}{c^2} \right) \right] = \frac{t_1}{c^2}.$$

$$E[C(t_1)C(t_2)] = c^2 E \left[B \left(\frac{t_1}{c^2} \right) B \left(\frac{t_2}{c^2} \right) \right] = t_1 = \min(t_1, t_2).$$

Continuity Properties of Brownian Motion

One of the most fundamental properties of Brownian motion is that its sample paths are **continuous functions of time with probability one**. Symbolically,

$$\mathbb{P}(B(t) \text{ is continuous in } t \in [0,1]) = 1.$$

But what does “continuous with probability 1” really mean?

A first naïve attempt might be:

For every $t \in [0, 1]$ and every $\varepsilon > 0$,

$$\mathbb{P}(|B(s) - B(t)| > \varepsilon) \rightarrow 0 \text{ as } s \rightarrow t.$$

This is *pointwise continuity in probability*. However, this condition is too weak. It does **not** characterize continuity of sample paths.

Indeed, the **Poisson process** satisfies exactly the same property!

Let $N(t)$ be a Poisson process with rate λ , and let $0 < \varepsilon < 1$. Then

$$\begin{aligned}\mathbb{P}(|N(s) - N(t)| > \varepsilon) &= \mathbb{P}(|N(s) - N(t)| > 0) \\ &= 1 - \mathbb{P}(N(s) = N(t)) \\ &= 1 - e^{-\lambda|t-s|} \\ &\rightarrow 0 \text{ as } s \rightarrow t.\end{aligned}$$

Yet we know that the Poisson process is **not** continuous: it has jump discontinuities at its arrival times.

So “continuity in probability” is not strong enough.

We need a stronger, pathwise statement.

Correct Continuity Statement

The correct formulation is:

$$\mathbb{P}\left(\lim_{s \rightarrow t} B(s) = B(t) \text{ for all } t \in [0,1]\right) = 1$$

With probability one, a single Brownian path is a continuous function of $t \in [0,1]$.

Equivalently,

$$\mathbb{P}\left(\exists t \in [0,1] \text{ such that } \lim_{s \rightarrow t} B(s) \neq B(t)\right) = 0.$$

So the probability that *any* discontinuity occurs anywhere on $[0,1]$ is zero.

More details involves deeper analysis and measure theory,

